Assignment 2 (Cloud Services & Security)

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1. Create a docker image with nginx as base image . The image should copy index.html within nginx container . Run the container & verify if your index.html was created in nginx container . Access the index.html from browser

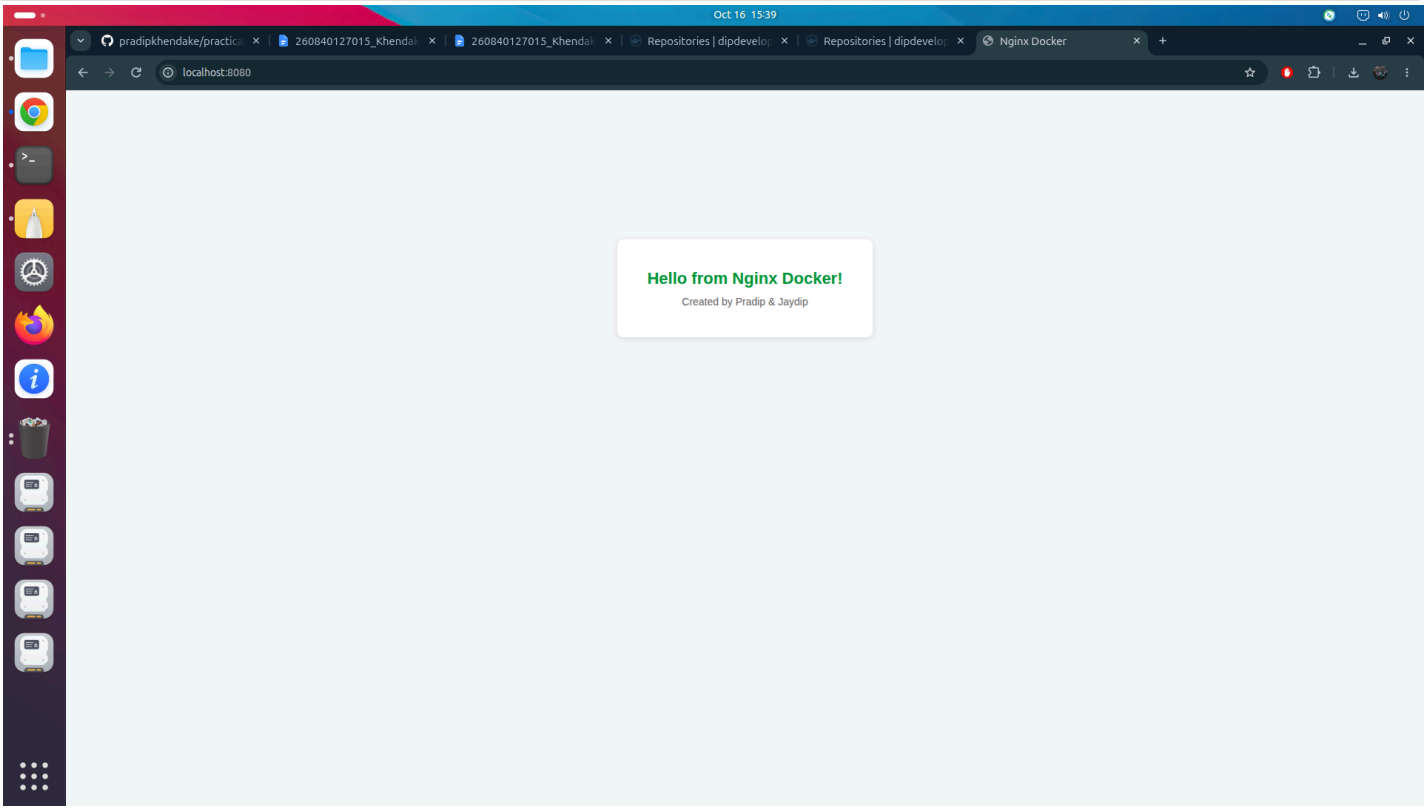
```
pradip@ProBook:~/git_practical$ cd
pradip@ProBook:~$ mkdir nginx-docker-demo
pradip@ProBook:~$ cd nginx-docker-demo
pradip@ProBook:~/nginx-docker-demo$ nano index.html
pradip@ProBook:~/nginx-docker-demo$ nano Dockerfile
```

```
<!DOCTYPE html>
<html>
<head>
    <title>Nginx Docker</title>
    <style>
  Terminal  body { font-family: sans-serif; text-align: center; padding-top: 20vh; background: #f4f6f8; color: #333; }
        .card { background: white; padding: 40px; border-radius: 8px; display: inline-block; box-shadow: 0 2px 8px rgba
        h1 { color: #009639; font-size: 22px; margin: 0 0 10px; }
        p { color: #666; margin: 0; font-size: 14px; }
    </style>
</head>
<body>
    <div class="card">
        <h1>Hello from Nginx Docker!</h1>
        <p>Created by Pradip & Jaydip</p>
    </div>
</body>
</html>
```

```
  GNU nano 7.2                                                    Dockerfile
FROM nginx:latest

COPY index.html /usr/share/nginx/html/index.html
```

```
pradip@ProBook:~/nginx-docker-demo$ docker build -t dipdevelopers/my-nginx-image .
[+] Building 2.9s (7/7) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 105B
 => [internal] load metadata for docker.io/library/nginx:latest
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [internal] load build context
 => => transferring context: 32B
 => [1/2] FROM docker.io/library/nginx:latest@sha256:332d0beea97cff8c0d5d27976c0778a4b00345b9d74c2feaaf8efde548314948
 => => resolve docker.io/library/nginx:latest@sha256:332d0beea97cff8c0d5d27976c0778a4b00345b9d74c2feaaf8efde548314948
 => CACHED [2/2] COPY index.html /usr/share/nginx/html/index.html
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:aaf25366663f4789947377e8feb9953fa57e2c4a43553c74ce6fe72e43f6f3b2
 => => exporting config sha256:50d559dfcca861cbf6dfe3df911550932432c75fd35f322b780f9783a26d41ab
 => => exporting attestation manifest sha256:ca0d57cd0a998b5ccf258cdfbd2a210cf27b54e1eaf4e321edf88620d8b50948
 => => exporting manifest list sha256:e86d3a85d621b8ab18a891fc8d0f4a4a7151a6ff7f1d64a035f3e7e76e514e21
 => => naming to docker.io/dipdevelopers/my-nginx-image:latest
 => => unpacking to docker.io/dipdevelopers/my-nginx-image:latest
pradip@ProBook:~/nginx-docker-demo$ docker run -d --name my-nginx-container -p 8080:80 dipdevelopers/my-nginx-image
038a2dfa559cc08c9f0850145cef64774d1f7bc33eb5ac0f00a2252b7c9d9050
pradip@ProBook:~/nginx-docker-demo$ docker exec -it my-nginx-container ls -la /usr/share/nginx/html
total 16
drwxr-xr-x 1 root root 4096 Oct 16 09:39 .
drwxr-xr-x 1 root root 4096 Oct 14 18:41 ..
-rw-r--r-- 1 root root  497 Oct  1 17:19 50x.html
-rw-rw-r-- 1 root root  194 Oct 16 09:37 index.html
```



2. Create a Docker file to install curl on alpine . Buid & Run the container . The curl command should access google .com on running the container

```
pradip@ProBook:~/nginx-docker-demo$ cd
pradip@ProBook:~$ mkdir alpine-curl-demo
pradip@ProBook:~$ cd alpine-curl-demo
pradip@ProBook:~/alpine-curl-demo$ nano Dockerfile
```

pradip@ProBook: ~/alpine-curl-demo

```
 GNU nano 7.2                                Dockerfile
FROM alpine:latest
RUN apk update && apk add curl

CMD ["curl", "https://www.google.com"]
```

```
pradip@ProBook:~/alpine-curl-demo$ docker build -t my-alpine-curl .
[+] Building 14.8s (7/7) FINISHED                                docker:default
 => [internal] load build definition from Dockerfile                      0.4s
 => => transferring dockerfile: 127B                                      0.0s
 => [internal] load metadata for docker.io/library/alpine:latest          3.6s
 => [auth] library/alpine:pull token for registry-1.docker.io             0.0s
 => [internal] load .dockerignore                                         0.3s
 => => transferring context: 2B                                           0.0s
 => [1/2] FROM docker.io/library/alpine:latest@sha256:294b683cb724975bec92580e1e685  0.4s
 => => resolve docker.io/library/alpine:latest@sha256:294b683cb724975bec92580e1e685  0.2s
 => [2/2] RUN apk update && apk add curl                                  6.3s
 => exporting to image                                                    2.8s
 => => exporting layers                                                   1.5s
 => => exporting manifest sha256:7babfdc177f5d6452ea28aeff2ac576c437344bf1b7b660505  0.2s
 => => exporting config sha256:80d2e179d1e10148e92f5b7ba4f9178c04bd0981a6d5eec2c0f7  0.2s
 => => exporting attestation manifest sha256:6d1aa5ab7a123795ce0166ceac2c56f2f39e34  0.2s
 => => exporting manifest list sha256:9cb9073a6bba5ef84ec7fe7e0565f5899d8019f269021  0.2s
 => => naming to docker.io/library/my-alpine-curl:latest                  0.0s
 => => unpacking to docker.io/library/my-alpine-curl:latest               0.3s
pradip@ProBook:~/alpine-curl-demo$ docker run my-alpine-curl
```

```
pradip@ProBook:~/alpine-curl-demo$ docker run my-alpine-curl
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0                              0<!doctype html><html itemscope="" itemtype="http://schema.org/WebPage" lang="en-IN"><head><meta content=
"text/html; charset=UTF-8" http-equiv="Content-Type"><meta content="/images/branding/googleg/1x/googleg_standard_color_128dp.png" itemprop="image"><title>Google</title><script nonce
="SRk4N_N88fkXxHHdJFU4mQ">(function(){var _g={kEI:'a7HRatitDfOK5OUP6uC8qAM',kEXPI:'0,4318858,16105,344796,301150,5232527,79,180,164,197,44,5990724,9,56200353,65168,138346,37805,1487
47,3050,2,48769,28575,17762,32563,6437,9742,2646,4521,20505,15108,2,1,1515,3355,2,29423,5089,3026,25,771,4081,4,31288,1250,5337,2891,2,12,16,28227,5,964,2,147,34657,2,6335,8497,4231
,1601,2,18519,6293,10880,264,12284,2676,10681,9282,2964,8,7179,3933,4,21020331,3,10055,3,17968,3,5865,2327,6504002,6,8302,8374,3,4225,3,1567,223,3,2312,570,3,4324,3,2883,183,3762,94
8,396,1117,2269,144,65167,786,1889803,711596,14927717,1503667,1,529043,47949,3,215328,417855,1,8246,2,52893,280742,3,12,1,2305,474,6777,7,7,16852,8,1239,1109,4895,5930,4388,5,13,5,4
245,22904,5,7854,4,4988,5,4025,2,1,8,6832,2634,5302,3396,1048,816,1734,7889,955,2139,4,4585,911,5834,1530,15059,2,1,8,15829,15,19,19949,4,1480,8330,8673,4,26,278,757,5,58,4,303,5636
,2048,4,421,4,3460,2,1,4862,221,5,14,5567,5,12131,1,1,3,14,1,4,1,3744,5,4786,1245,2,1,16,252,1707,846,974,4,131,4,480,4,2855,4,2103,2,453,5,202,4,11964,4,933,4,69,5,7851,2879,5,1253
,5,653,4,3797,276,4,191,3876,1,2711,1,563,5,1144,4,3304,2830,4,10438,2,466,2228,1071,5,2750,4,1566,203,4,2828,1712,2,281,4,2047,4,1478,4,1122,2,1627,232,4,2698,4,1410,6023,5,2587,7,
143,4570,4,675,1289,4,571,236,1084,3,2,2,2,3136,3,3626,1119,255,5,66,4,2702,1,2,6515,2704,3,2,2,2,78,1333,2395,242,73,9,4,2415,546,5044,1678,4047,4,512,4,1410,18,28,2210,5051,344,51
99,5,240,179,244,2,4,8211,6,390,189,5,884,2622,958,4,306,914,4,526,1311,4,151,291,4,10,1,1077,490,1774,5,1191,4,2096,2259,1143,4,167,742,4,1,64,4,154,98,4,497,4327,963,1548,1316,377
8,2394,4,996,1,2,19,242,53,4,1033,3543,467,1,363,4,1,548,597,654,4,2628,187,1461,5,13,4,4096,5,1558,4,78,1258,10,166,5,1817,1112,4,997,4,1,254,4,583,1406,5,30,69,1,2,2,162,4,97,2055
,574,1,344,5,2,6,272,5,228,5,1132,874,9,1465,1658,4,209,209,435,4,1873,4,5099,4,560,5,111,59,779,5,994,1088,4,1,1157,251,4,2,1911,2029,702,4,216,1184,524,16,4,1,943,294,265,151,7,89
4,4,1187,87,4,583,85,4,4,540,5,1652,2,1,2424,11,2155,3,2,2,2,511,207,65,4,3504,262,1,220,49,270,4,557,4,1,448,4,1,99,5,824,1242,4,126,4,99,4,4,169,673,58,1558,291,4,13,2,199,1796,13
15,1,2,549,3,106,5,138,177,4,756,4,3877,3,66,1854,4,8,462,712,56,150,4,10,201,476,1,2,990,2,185,4,584,4,745,904,5,251,5,846,63,3,2,2,2,322,4,31,4,390,1456,4,394,4,326,1702,355,4,556
,216,116,789,717,3,2,2767,3,2,2,2,391,4,395,4,90,4,240,4,174,95,591,4,59,4,2,372,1237,79,5,573,280,3,2,2,2,2770,59,4,11,921,29,4,1,209,4,505,95,4,3678,207,4,1503,4,1062,37,3,2,2,2,3
94,4,693,530,22,175,4,341,4,340,1,2,258,739,4,51,10,4,865,1205,503,723,4,395,93,535,1941,10,46,196,501,3,2,2,2,336,33,80,1,516,1053,73,5,682,4,84,5,127,1992,4,1,57,86,816,1162,142,1
5,610,1,474,3,2,2,2,1287,4,462,4,556,4,663,4,482,302,118,285,203,2,1,221,6,43,1425,1,870,4,187,4,162,1812,957,3,938,191,19,492,5,462,2,1,4,2,147,54,6,1,102,4,1341,4,422,319,5,164,4,
806,152,251,5,207,1,2,1331,4,14,8,1024,7,63,4,357,1663,133,216,3,9,4,11,18,1011,246,3,2,28,209,2,1881,5,383,3,2,2,2,1692,1,4,154,40,12,620,512,284,1863,2243,631,124,2797,1297,4,2,52
4,65,42,11,1,675,4,379,926,6,2188,338,285,3,2,2,2,722,2751,4,1715,2199,231,3,3,2,2222,2,299,7,113,4,193,3,27,579,519,1,874,579,488,14,61,3622,4,946,192,112,13,3921,3,2,2,2,442,759,3
,2,2,2,247,23,5,1443,3,3071,536,4,677,599,990,4,31,253,301,4,4,1,548,33,3,2,2,2,4948,47,2,5029,1,2,1093915,12,1,109737,8,14',kBL:'wpOf',kOPI:89978449};(function(){var a;((a=window.g
oogle)==null?0:a.stvsc)?google.kEI=_g.kEI:window.google=_g;}).call(this);})();(function(){google.sn='webhp';google.kHL='en-IN';google.usb=true;})();(function(){
var h=this||self;var k=/^|;/;function l(a,c){var d=a.search(k);a:{var b=0;for(var f=c.length;(b=a.indexOf(c,b))>=0&&b<d;){var e=a.charCodeAt(b-1);if(e==38||e==63)if(e=a.charCodeAt(b
+f),!e||e==61||e==38||e==35)break a;b+=f+1}b=-1}if(b<0)return null;f=a.indexOf("&",b);if(f<0||f>d)f=d;b+=c.length+1;return decodeURIComponent(a.slice(b,f!==-1?f:0).replace(/\+/g," "
))};function m(){return window.google&&window.google.kOPI||null};var n;function p(a){for(var c;a&&(!a.getAttribute||!(c=a.getAttribute("eid")));)a=a.parentNode;return c||n}function
q(a){for(var c=null;a&&(!a.getAttribute||!(c=a.getAttribute("leid")));)a=a.parentNode;return c}
function t(a,c,d,b,f){var e="";c.search("&ei=")===-1&&(e="&ei="+p(b),c.search("&lei=")===-1&&(b=q(b))&&(e+="&lei="+b));var r=c.search("&cshid=")===-1&&a!=="slh";var g=g===void 0?win
dow.location:g;b="&zx="+Date.now().toString();h._cshid&&r&&(b+="&cshid="+h._cshid);(d=d())&&(b+="&opi="+d);(d=l(g.search,"gemcid"))&&(b+="&gemcid="+d);(d=l(g.search,"gemrid"))&&(b+=
"&gemrid="+d);(g=l(g.search,"gemhist"))&&(b+="&gemhist="+g);return"/"+(f||"gen_204")+"?atyp=i&ct="+String(a)+"&cad="+(c+e+b)};n=google.kEI;google.getEI=p;google.getLEI=q;google.ml=f
unction(){return null};google.log=function(a,c,d,b,f,e){e=e===void 0?m:e;d||(d=t(a,c,e,b,f));a=d;/^http:/i.test(a)&&window.location.protocol==="https:"&&(google.ml&&google.ml(Error(
"a"),!1,{src:a,glmm:1}),a="");if(d=a)try{window.navigator.sendBeacon(d,"")}catch(r){}};google.logUrl=function(a,c){c=c===void 0?m:c;return t("",a,c)};}).call(this);(function(){googl
e.y={};google.sy={};function e(a,b,d){if(a)var c=a.id;else{do c=Math.random();while(d[c])}d[c]=[a,b]}var f;(f=google).x||(f.x=function(a,b){e(a,b,google.y)});var g;(g=google).sx||(g
.sx=function(a,b){e(a,b,google.sy)});google.bx=!1;var h;(h=google).lx||(h.lx=function(){});var k=[],l;(l=google).fce||(l.fce=function(a,b,d,c){k.push([a,b,d,c])});google.qce=k;googl
```

Assignment 3 -

1. Create a new git repo, add a Hello.java file, complete the clone-add-commit-push steps. Then
modify the code and repeat.

2. Do the above using git init.

```
pradip@ProBook:~$ cd hello/
pradip@ProBook:~/hello$ ls
Hello.java
pradip@ProBook:~/hello$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/pradip/hello/.git/
```

```
pradip@ProBook:~/hello$ git add *
pradip@ProBook:~/hello$ git commit -m "Commit from Main branch"
[master (root-commit) 31bae57] Commit from Main branch
 1 file changed, 11 insertions(+)
 create mode 100644 Hello.java
pradip@ProBook:~/hello$ git branch
* master
pradip@ProBook:~/hello$ git branch -m master main
pradip@ProBook:~/hello$ git branch
* main
pradip@ProBook:~/hello$ git checkout -b feature_branch
Switched to a new branch 'feature_branch'
pradip@ProBook:~/hello$ git branch
* feature_branch
  main
pradip@ProBook:~/hello$ ls
Hello.java
pradip@ProBook:~/hello$ nano Hello.java
```

```
  GNU nano 7.2                          Hello.java *
import java.util.Scanner;

public class Hello {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Hello, Welcome friends to our World !!");

        scanner.close();
    }
}
```

3. Demonstrate the branch-merge feature for the above repo.

```
pradip@ProBook:~/hello$ git branch
  feature_branch
* main
pradip@ProBook:~/hello$ git checkout feature_branch
M       Hello.java
Switched to branch 'feature_branch'
```

```
pradip@ProBook:~/hello$ cat Hello.java
import java.util.Scanner;

public class Hello {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Hello, Welcome friends to our World from feature branch !!");

        scanner.close();
    }
}
```

```
pradip@ProBook:~/hello$ git checkout main
Switched to branch 'main'
pradip@ProBook:~/hello$ nano Hello.java
pradip@ProBook:~/hello$ git add *
pradip@ProBook:~/hello$ git commit -m "Commit from Main_branch"
[main 533cb3e] Commit from Main_branch
 1 file changed, 1 insertion(+), 1 deletion(-)
pradip@ProBook:~/hello$ git merge feature_branch
Auto-merging Hello.java
CONFLICT (content): Merge conflict in Hello.java
Automatic merge failed; fix conflicts and then commit the result.
```

```
  GNU nano 7.2                                Hello.java
import java.util.Scanner;

public class Hello {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

<<<<<<< HEAD
        System.out.print("Hello, Welcome friends !! from main branch");
=======
        System.out.print("Hello, Welcome friends to our World from feature branch !!");
>>>>>>> feature_branch

        scanner.close();
    }
}
```

4. Create a Java application to accept a number and print its factorial. Build a Docker image and name it as my-java-app. Run it as a Docker container.

```
pradip@ProBook:~/hello$ cd
pradip@ProBook:~$ mkdir java_code
pradip@ProBook:~$ cd java_code/
pradip@ProBook:~/java_code$ nano Factorial.java
pradip@ProBook:~/java_code$ nano Dockerfile
```

Terminal

```
import java.util.Scanner;

public class Factorial {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int num = scanner.nextInt();

        long factorial = 1;
        for (int i = 1; i <= num; i++) {
            factorial *= i;
        }

        System.out.println("Factorial of " + num + " is: " + factorial);
        scanner.close();
    }
}
~
~
~
~
~
~
"~/java_code/Factorial.java" 17L, 461B                    1,1           A
```

Open — Dockerfile — ~/java_code — Ln 1, Col 1

```
1 FROM eclipse-temurin:17-jdk
2 WORKDIR /app
3 COPY Factorial.java .
4 RUN javac Factorial.java
5 CMD ["java", "Factorial"]
```

```
pradip@ProBook:~/java_code$ docker build -t my-java-app .
[+] Building 40.2s (10/10) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 151B
 => [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk
 => [auth] library/eclipse-temurin:pull token for registry-1.docker.io
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/4] FROM docker.io/library/eclipse-temurin:17-jdk@sha256:fe947d993c32dd9cf651a4353a6ed723c2370b3a9d034ea4cab4775404ea531a
 => => resolve docker.io/library/eclipse-temurin:17-jdk@sha256:fe947d993c32dd9cf651a4353a6ed723c2370b3a9d034ea4cab4775404ea531a
 => => sha256:80e37595eab7b7312576d0e0b537134e6e7ee99a70d194c157fe7c4bfe3d5a1f 2.46kB / 2.46kB
 => => sha256:ec856601565e5587d3652caae344701db794fa9dca67bb10a69d27626ba9ae5b 159B / 159B
 => => sha256:7b5e170ba38382bc2458627c781bfed5358ff04835eccfe22f7f9ffee8d2bfb2 145.83MB / 145.83MB
 => => sha256:1d971d0370912eae994f900a5f4a0dd7ec82f60e615ba386efc27353d85337d6 24.06MB / 24.06MB
 => => sha256:d9d511d19a19ebdeda520e0752f184d457905c7a310df5b748d133a51274ecd4 391B / 391B
 => => sha256:435bf5c20694c99dd5a9877e58e24b26a2207fcf8c692a3c3ce3c867770a5b41 42.15MB / 42.15MB
 => => extracting sha256:435bf5c20694c99dd5a9877e58e24b26a2207fcf8c692a3c3ce3c867770a5b41
 => => extracting sha256:d9d511d19a19ebdeda520e0752f184d457905c7a310df5b748d133a51274ecd4
 => => extracting sha256:1d971d0370912eae994f900a5f4a0dd7ec82f60e615ba386efc27353d85337d6
 => => extracting sha256:7b5e170ba38382bc2458627c781bfed5358ff04835eccfe22f7f9ffee8d2bfb2
 => => extracting sha256:ec856601565e5587d3652caae344701db794fa9dca67bb10a69d27626ba9ae5b
 => => extracting sha256:80e37595eab7b7312576d0e0b537134e6e7ee99a70d194c157fe7c4bfe3d5a1f
 => [internal] load build context
 => => transferring context: 504B
 => [2/4] WORKDIR /app
 => [3/4] COPY Factorial.java .
 => [4/4] RUN javac Factorial.java
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:9434421dacd5e7a30225b029fe2c7d42677b09a88002ebcbc0c6fdc0a0dcc27d
 => => exporting config sha256:63f41bc2555fd9bedf4d2df46b8d9a2cf2562af47d86e0888dc176a426fc6225
 => => exporting attestation manifest sha256:9fd6656ebcaa089bc9f6dff679c0f1f8244d6ab2a4b79f6f0877a7eb1ab5ba92
 => => exporting manifest list sha256:b1cabd260c78942c19d804ffc994625e8b53c5bf93b8387b5ad76ea0fd2e18af
 => => naming to docker.io/library/my-java-app:latest
 => => unpacking to docker.io/library/my-java-app:latest
```

```
pradip@ProBook:~/java_code$ docker run -it my-java-app
Enter a number: 5
Factorial of 5 is: 120
pradip@ProBook:~/java_code$ docker run -it my-java-app
Enter a number: 3
Factorial of 3 is: 6
```

5. Push the above image to DockerHub.

```
pradip@ProBook:~/java_code$ docker tag my-java-app dipdevelopers/my-java-app:latest
pradip@ProBook:~/java_code$ docker push dipdevelopers/my-java-app:latest
The push refers to repository [docker.io/dipdevelopers/my-java-app]
44136fa355b3: Mounted from dipdevelopers/my-hello-python
435bf5c20694: Pushed
d9d511d19a19: Pushed
0541cf7dd420: Pushed
1d971d037091: Pushed
7b5e170ba383: Pushed
ec856601565e: Pushed
80e37595eab7: Pushed
54317a7aecc3: Pushed
963c293462ea: Pushed
c225c8f0da5d: Pushed
latest: digest: sha256:b1cabd260c78942c19d804ffc994625e8b53c5bf93b8387b5ad76ea0fd2e18af size: 856
```

6. Now modify code to accept 3 numbers and find which is the largest. Create a new image and run a container.

```
pradip@ProBook:~/java_code$ nano Largest.java
pradip@ProBook:~/java_code$ nano Dockerfile
```

```
 GNU nano 7.2                                Largest.java
import java.util.Scanner;

public class Largest {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter three numbers separated by spaces: ");
        int a = scanner.nextInt();
        int b = scanner.nextInt();
        int c = scanner.nextInt();

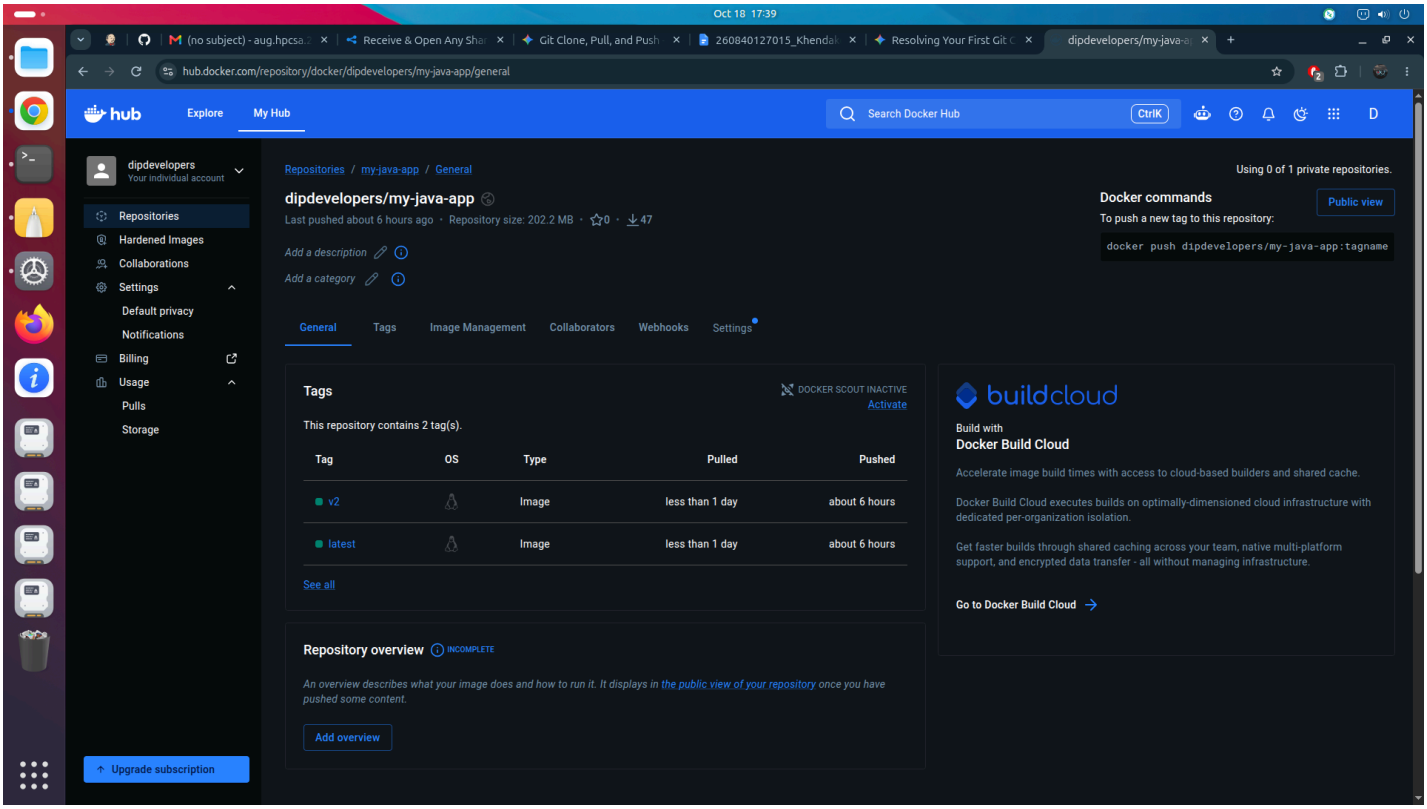
        int largest = (a >= b && a >= c) ? a : (b >= c ? b : c);
        System.out.println("The largest number is: " + largest);
        scanner.close();
    }
}
```

```
 GNU nano 7.2                                Dockerfile
FROM eclipse-temurin:17-jdk
WORKDIR /app
COPY Largest.java .
RUN javac Largest.java
CMD ["java", "Largest"]
```

```
pradip@ProBook:~/java_code$ docker build -t my-java-app:v2 .
[+] Building 9.6s (10/10) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 145B
 => [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk
 => [auth] library/eclipse-temurin:pull token for registry-1.docker.io
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/4] FROM docker.io/library/eclipse-temurin:17-jdk@sha256:fe947d993c32dd9cf651a4353a6ed723c2370b3a9d034ea4cab4775404ea531a
 => => resolve docker.io/library/eclipse-temurin:17-jdk@sha256:fe947d993c32dd9cf651a4353a6ed723c2370b3a9d034ea4cab4775404ea531a
 => [internal] load build context
 => => transferring context: 526B
 => CACHED [2/4] WORKDIR /app
 => [3/4] COPY Largest.java .
 => [4/4] RUN javac Largest.java
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:eb9ddcd86ba262ab195f3c1b7019a6a74b721d64e893c29e350d7b9d0d40dd19
 => => exporting config sha256:e691ebbb711fd1ee9809ddbbaad0f69ef077b0d10b017253b2134e350d6f7847
 => => exporting attestation manifest sha256:b4d674eaea17b1878b2f7917c7588f739e679d413d06495c12005a13ad1ca9ee
 => => exporting manifest list sha256:6e17a9402361129e7692f9f1ee01ed7d142c6af1d5519f373887e451c5352f25
 => => naming to docker.io/library/my-java-app:v2
 => => unpacking to docker.io/library/my-java-app:v2
pradip@ProBook:~/java_code$ docker run -it my-java-app:v2
Enter three numbers separated by spaces: 20 30 50
The largest number is: 50
```

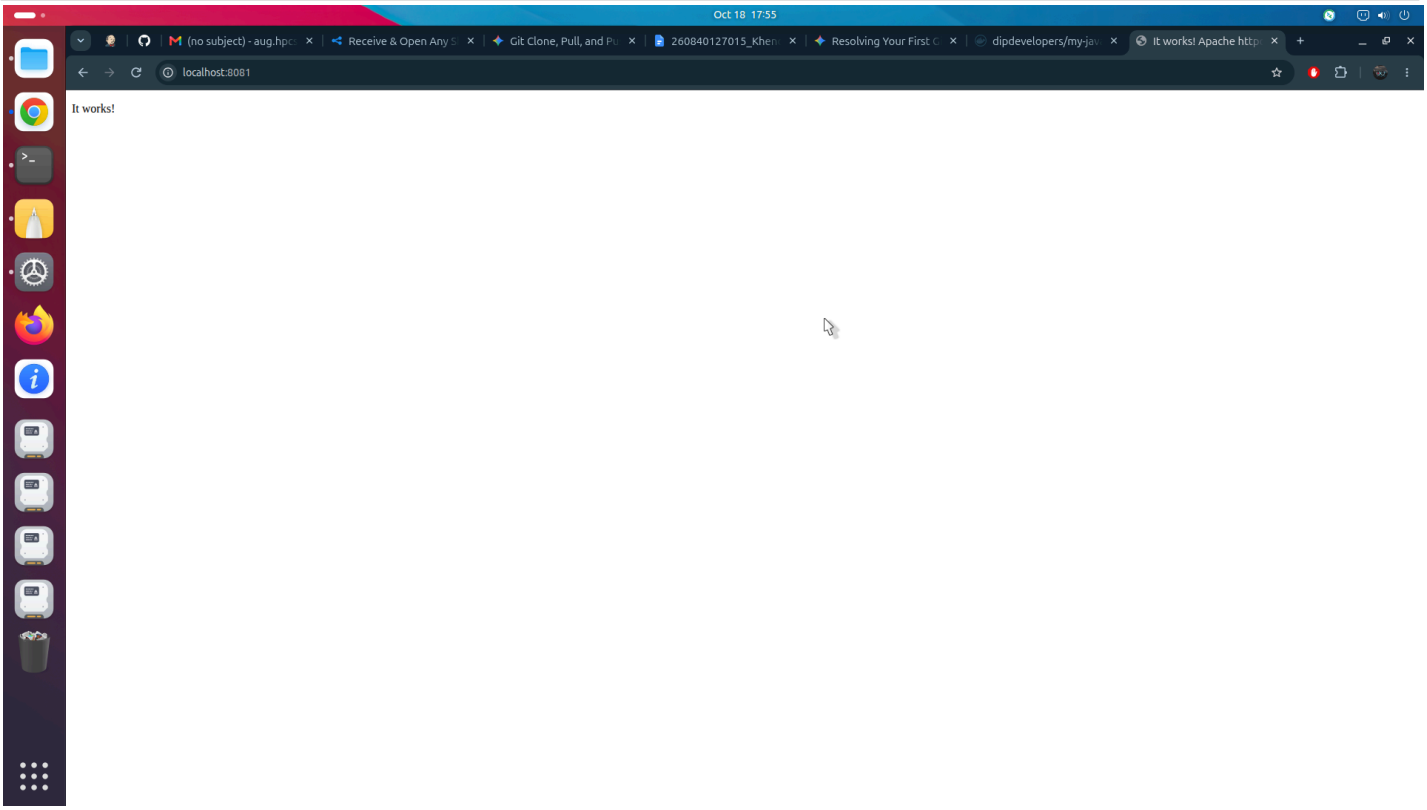
7. Push the new image as version 2.

```
pradip@ProBook:~/java_code$ docker tag my-java-app:v2 dipdevelopers/my-java-app:v2
pradip@ProBook:~/java_code$ docker push dipdevelopers/my-java-app:v2
The push refers to repository [docker.io/dipdevelopers/my-java-app]
d9d511d19a19: Layer already exists
36c0292e1cbf: Pushed
44136fa355b3: Already exists
435bf5c20694: Layer already exists
1d971d037091: Layer already exists
7b5e170ba383: Layer already exists
ec856601565e: Layer already exists
80e37595eab7: Layer already exists
54317a7aecc3: Layer already exists
52bad6cb9ba0: Pushed
f2d925e3d7c3: Pushed
v2: digest: sha256:6e17a9402361129e7692f9f1ee01ed7d142c6af1d5519f373887e451c5352f25 size: 856
```



8. Run an Apache web server container and view its home page in the browser.

```
1dd09ede7236
pradip@ProBook:~/java_code$ docker run -d -p 8081:80 --name my-apache-server httpd
24f196b885f40946fbd5416d200bcd834772b0f67c3ef77b675aaa5116055be1
pradip@ProBook:~/java_code$ docker ps
CONTAINER ID   IMAGE     COMMAND              CREATED         STATUS         PORTS                                   NAMES
24f196b885f4   httpd     "httpd-foreground"   2 minutes ago   Up 2 minutes   0.0.0.0:8081->80/tcp, [::]:8081->80/tcp   my-apache-server
pradip@ProBook:~/java_code$
```


Oct 18 17:55
localhost:8081
It works!